SCm CMB Angular Power Spectrum via Phonon-Modulated Primordial Integration

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Abstract

We derive the full SCm CMB angular power spectrum by integrating the phonon-modulated primordial power spectrum over k-space:

$$C_\ell^{\mathrm{SCm}} = \frac{2}{\pi} \int dk\, k^2\, P_{\mathrm{SCm}}(k)\, |\Delta_T^{\mathrm{SCm}}(\ell, k)|^2$$

where the SCm-modified primordial spectrum is:

$$P_{\mathrm{SCm}}(k) = A_s \left(\frac{k}{k_{\mathrm{piv}}}\right)^{n_s - 1} [1 + \Phi \cdot S_{26} \cdot (2R - 1)]$$

This extends the simplified CMBAngularPowerCalculator (grok_100_equations_module.py) to a full k-space integration with Sachs-Wolfe, acoustic, and Silk damping transfer functions, all modulated by the SCm phonon linewidth Γ.

$§$1 SCm-Modified Primordial Spectrum

The standard nearly scale-invariant spectrum $P(k) = A_s(k/k_{\mathrm{piv}})^{n_s - 1}$ is boosted by the SCm buoyancy factor:

$$P_{\mathrm{SCm}}(k) = P(k) \cdot \left[1 + \Phi_{1.25\,\mathrm{THz}} \cdot S_{26}^{(3)}([\mathrm{SSq}]) \cdot (2R - 1)\right]$$

with $A_s = 2.1 \times 10^{-9}$ (Planck 2018), $n_s = 0.965$, $k_{\mathrm{piv}} = 0.05\ \mathrm{Mpc}^{-1}$, $\Phi_0 = 10^{20}$, $S_{26} \approx 1.86$.

$§$2 Temperature Transfer Function

$$\Delta_T^{\mathrm{SCm}}(\ell, k) = \Delta_T^{\mathrm{SW}}(\ell, k) + 0.6 \cdot \Delta_T^{\mathrm{acoustic}}(\ell, k)$$

$§$2.1 Sachs-Wolfe Term

$$\Delta_T^{\rm SW} = \frac{1}{3}\exp\left(-\frac{x^2}{2}\right), \quad x = \frac{k \cdot r_*}{\ell + 0.5}$$

where $r_* \approx 14{,}000$ Mpc is the comoving sound horizon at last scattering.

$§$2.2 Acoustic Oscillation Term

$$\Delta_T^{\rm acoustic} = \cos(x) \cdot \exp\left(-\frac{k^2}{k_D^2}\right)$$

with Silk damping scale $k_D \approx 0.15$ Mpc^{-1}.

$§$3 Full Power Spectrum Integral

$$C_\ell^{\rm SCm} = \frac{2}{\pi}\int_{k_{\rm min}}^{k_{\rm max}} dk\, k^2\, P_{\rm SCm}(k)\, |\Delta_T^{\rm SCm}(\ell, k)|^2 \cdot (T_0 \times 10^6)^2$$

in units of μK^2.

$§$4 Acoustic Peak Structure

Peak	Multipole ℓ	$C_\ell^{\rm SCm}$ (μK^2)	Physical Origin
1st	~ 220	Dominant	SW + first compression
2nd	~ 546	Subdominant	First rarefaction
3rd	~ 800	Tertiary	Second compression

The SCm boost factor $1+\Phi S_{26}(2R-1)$ uniformly amplifies all peaks, preserving the observed peak ratio structure while providing a physical origin via vacuum phonon resonance.

$§$5 Advantages over Standard Treatment

Aspect	Standard ΛCDM	SCm Phonon
Primordial spectrum	Inflaton slow-roll	SCm vacuum buoyancy
Free parameters	A_s, n_s, r	Φ_0, Γ, [SSq]
Physical mechanism	Quantum fluctuations	Phonon resonance at 1.25 THz
Peak amplification	Static A_s	Dynamic $\Phi(\Gamma)$
Testability	CMB polarization	THz phonon linewidth

References

- PAPER_1093: SCm CMB Temperature Fluctuation
- PAPER_1094: CMB Buoyancy Sector Lagrangian
- PAPER_202: UQFF Reionization / BBN / Recombination
- PAPER_199: $F_{U,Bi,i}$ Taxonomy Part 2 — Cosmological Dark Sector